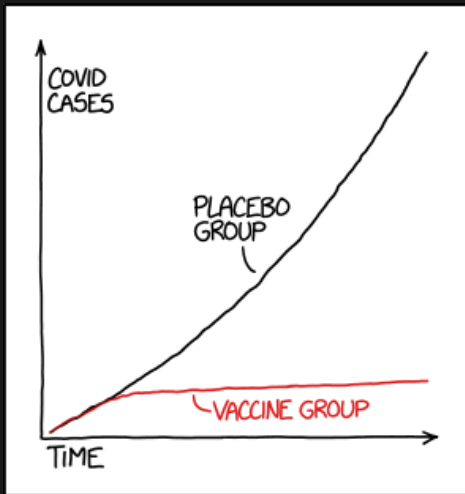


Towards Modern Statistical Testing

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Lecture 02.2 (v2.0.2)

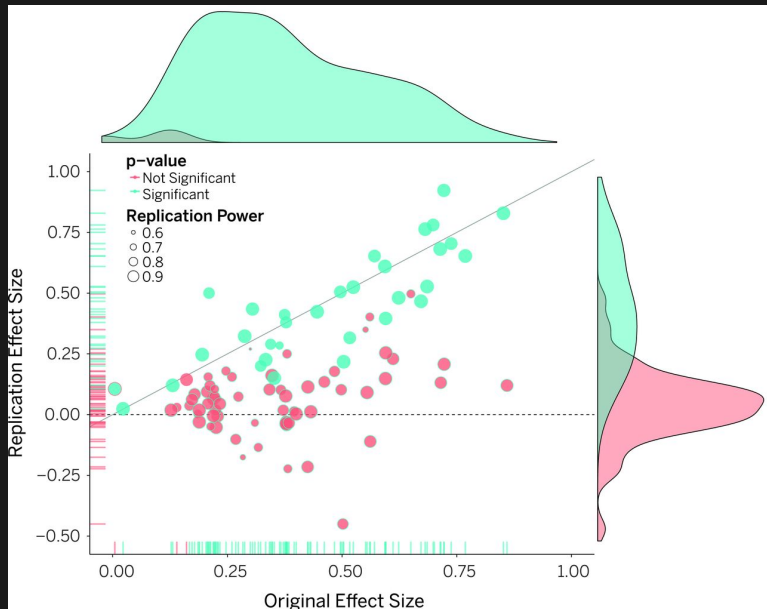


STATISTICS TIP: ALWAYS TRY TO GET DATA THAT'S GOOD ENOUGH THAT YOU DON'T NEED TO DO STATISTICS ON IT

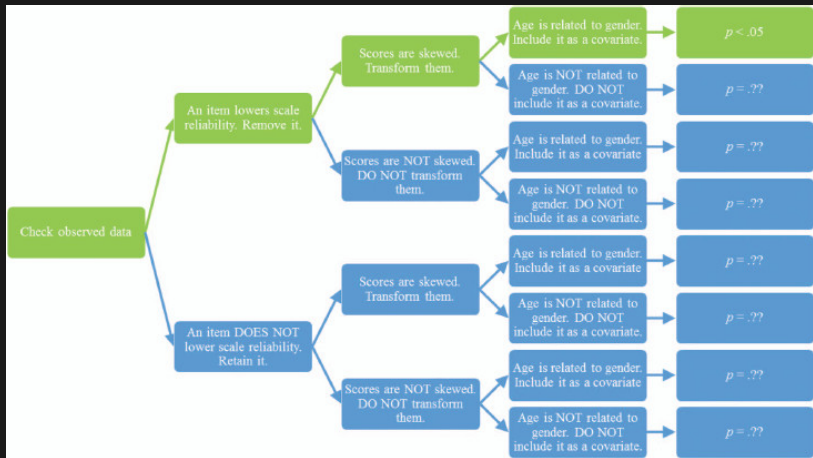
Signposting

- ▶ This lecture covers three main topics:
 - ▶ Classical testing (and when its still ok to use it)
 - ▶ Modern testing (and how to use it well)
 - ▶ General Cross Validation (and why you should always do it)

The replication crisis



The garden of forking paths



Questions

- ▶ Why might statistical testing have a bad reputation?
- ▶ What is the point of testing in large-scale data science?
- ▶ What is Monte-Carlo testing?
- ▶ What does(n't) Cross Validation save you from?

Null hypothesis test

- ▶ Given some data $\{y\}$:
 - ▶ Null Hypothesis **H0**: A statement is true about $\{y\}$.
 - ▶ Alternative Hypothesis **H1**: The statement is not true.
- ▶ We then compute a **test statistic** $T(\{y\})$ whose distribution is **computable under H0**.
 - ▶ By convention, large T is evidence against the null.
- ▶ Then compute p-value $p(T \geq T(\{y\}))$, the probability of observing a test statistic at least as large as that observed given H0 is true.
 - ▶ Example: H0: $\mathbb{E}(y) = \mu$ with $\mu = 0$. H1: $\mu \neq 0$.
 - ▶ This is **not model selection**. We favour H0 and must find evidence against it to accept H1.

Null hypothesis significance testing

- ▶ Hypothesis testing is asking: are my data consistent **with this hypothesis** when **using this measure**?
 - ▶ If you choose a silly hypothesis, testing will dutifully say “no”
 - ▶ If you use a weak measure, testing will dutifully say “yes”
 - ▶ Nothing is learned by this!
- ▶ The correct use of statistical testing is where:
 1. the **null hypothesis might plausibly be true**, or
 2. it might not be true, but you care how much **power the data has to reject the null**

When to use hypothesis testing

- ▶ Some valid use cases include:
 - ▶ To **rank hypotheses** by how much evidence there is against them
 - ▶ To obtain a **standardised scale** (0-1) for combining evidence
 - ▶ When **data are scarce**
- ▶ Also when testing plausible nulls, such as:
 - ▶ **validating simulations** with a known simulator;
 - ▶ **independence** or other non-parametric tests.
 - ▶ **broad null hypotheses**, such as testing a range of parameters.

Types of error

- ▶ The **p-value** defines *the probability that H_0 is true, but is rejected*.
- ▶ The **power of the test** is *the probability that H_0 is false but is accepted anyway*.
 - ▶ Low power situations are to be avoided: see e.g. Andrew Gelman's blog¹.
- ▶ Power is a surprisingly important problem because there are many *researcher degrees of freedom*.
 - ▶ so if power is low, we tend to find significant results anyway, through the (often unintentional) data and testing choice.

¹<https://andrewgelman.com/2018/02/18/low-power-replication-crisis-learned-since-2004-1984-1964/>

Types of error

Error notation

	H0 true	H0 false
H0 accepted	Correct	Type II error
H0 rejected	Type I error	Correct

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	H0 true	H0 false
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- Under the convention that $H0 = 0$ = “negative” case and $H1 = 1$ = “positive case”:

Alternative notation

	H0 true	H0 false
H0 accepted	True Negative	False Negative
H0 rejected	False Positive	True Positive

t-tests

- ▶ Can be one-tailed (**H0**: $\mu \leq \mu_0$) or two-tailed (**H0**: $\mu = \mu_0$)
- ▶ Assumes:
 - ▶ independence (note: paired tests are possible) and identically distributed
 - ▶ the **data are Normal**
 - ▶ the standard deviation is either known (t is then Normal) or estimated from the data (t is then t distributed).
- ▶ Used in regression, paired tests, etc.
- ▶ *NB Incomplete notes as this is a prerequisite!*

Chi squared test

- ▶ The χ^2 test is for categorical data comparing two variables.
- ▶ **H0**: No relationship between the variables; **H1** Some relationship between them.
- ▶ The **test statistic** for N datapoints from k classes, with x_i observations of type i , with expected value $m_i = Np_i$ where p_i is the expected probabilities, is (under the null):

$$X^2 = \sum_{i=1}^k \frac{(x_i - m_i)^2}{m_i} \sim \chi^2(k - 1)$$

- ▶ This is most often used for **contingency tables** though appears elsewhere.
- ▶ See also **Fishers exact test** for small samples.
- ▶ *NB Incomplete notes as this is a prerequisite!*

Other important tests

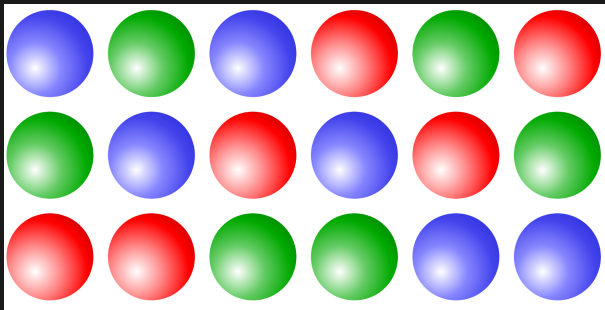
- ▶ Nonparametric tests:
 - ▶ **Mann-Whitney U or Wilcoxon rank sum** test: are two samples drawn from the same distribution? by comparing their ranks.
 - ▶ **Wilcoxon signed-rank** test - as rank sum test, for paired data.
 - ▶ **Kolmogorov-Smirnov** test - are two samples from the same distribution? by comparing the empirical cumulative distribution function.
- ▶ There are many online cookbooks which state exactly which circumstances each test should be used in. You should be able to use them.
- ▶ *NB Incomplete notes as this is a prerequisite!*

Resampling

- ▶ The main types of resampling tests include:
 - ▶ **jackknifing**, which is analysing subsets of data to estimate (variance of) parameter estimates
 - ▶ **bootstrapping**, which is resampling with replacement, to estimate (variance of) parameter estimates
 - ▶ **permutation**, which is resampling without replacement, to test a null hypothesis
 - ▶ **cross-validation**, which is analysing subsets of data to estimate out-of-sample prediction, for model performance
- ▶ Each of these methods can be applied to a wide variety of problems, and often requires thought to use appropriately.

Permutations

All permutations of three colors (each column is a permutation):



- Figure from Wikipedia². There are in general $n!$ permutations.

²https://upload.wikimedia.org/wikipedia/commons/4/4c/Permutations_RGB.svg / 32

Generating permutations

```
> set.seed(1)
> n = 5
> x = seq(0,20,length=n)
> x
[1] 0 5 10 15 20
> x[sample.int(n)]
[1] 5 20 15 10 0
> x[sample.int(n)]
[1] 20 15 5 10 0
```

Use of permutations in testing

- ▶ Consider the following general class of problem:
 - ▶ H_0 : y is independent of x .
 - ▶ H_1 : y is dependent on x .
- ▶ x may be continuous, categorical, etc and y may depend on a number of other things.
- ▶ A **permutation test** will:
 - ▶ resample x, y pairs **under H_0** ,
 - ▶ Construct a test statistic T ,
 - ▶ test if T is extreme in the real data, compared to the permutations?

Why permutations

- ▶ Asymptotically correct and distribution free. We only (!) have to assume **exchangeability**.
- ▶ Exchangeability of what?
 - ▶ what would be **equal if the null hypothesis is true**, and
 - ▶ would be **different if the alternative hypothesis is true**?
- ▶ The test is correct if we **maintain any true correlation structure**.
- ▶ For example, if the indices were originally correlated, permutation testing will fail.
 - ▶ as from e.g. a time-series.

Monte-Carlo testing

- ▶ There are in general $n!$ permutations. This is typically too many for $n > 20$.
- ▶ We instead choose N **random permutations** from all the possible ones.
- ▶ Monte-Carlo testing is an important subject in its own right.
- ▶ Its often possible to place guarantees on the p -value from very few samples.

Monte-Carlo test

- ▶ To conduct a Monte-Carlo test, we construct N random datasets and add our real dataset.
- ▶ We then ask, is the **real dataset an outlier** with respect to the random datasets?
- ▶ Specifically, the p-value for a test T applied to X (where large values are considered strange) is:

$$\frac{\text{Rank}(T(X); T(\{x_i\}))}{N + 1}$$

- ▶ where Rank simply counts the number of cases as large or larger.

Permutation testing summary

- ▶ **Distributional assumptions** are often invalid (regular tests)
- ▶ **Exchangeability assumptions** are often plausible (permutation tests)
- ▶ It is still possible to get misleading inference if the assumptions of a test don't hold

Model Selection

- ▶ Imagine that we have run two different inference procedures (models) on our data.
- ▶ We want to decide which of these gives the **best** description of the data.
 - ▶ (we might pretend we want to know which one is **right**...)
- ▶ Model selection formalises how to make this assessment.

General considerations

- ▶ To make Cross-Validation work, we need to be able to define our inference goal cleanly. Some scenarios:
 - ▶ **Same source, single datapoint**: Within a single datastream, how well can we predict the **next** point?
 - ▶ **Same source, segment of data**: Within a single datastream, how well could we predict everything that happens within an hour?
 - ▶ **New but understood source**: We have multiple datastreams, each of which might be different but all are generated by a similar process. How well can we predict a new such datasource?
 - ▶ **Unexpected source**: We have many classes of datastream. How well can we predict what would happen on a new class of datastream?

Problems with LOOCV

- ▶ We might worry that leaving out one datapoint at a time isn't enough:
 - ▶ **Cost**. It is straightforward to apply LOOCV to an arbitrary loss function, including a Likelihood. However, it can be costly.
 - ▶ **Quality**. LOOCV estimates of out-of-sample loss has high variance because each test datapoint using $n - 2$ of the **same training datapoints**. . .
 - ▶ Empirically, we often choose a different model on different data generated under the same distribution!
 - ▶ **Correlation**. Any correlation breaks LOOCV.

K-fold CV

- ▶ Naive **k-fold CV** addresses the first issue by creating a **bias-variance tradeoff**: we introduce a bias (towards simpler models) but also significantly reduce the variance of the MSE estimation.
- ▶ More complicated sampling in k-fold settings can also address correlation.
- ▶ **Split** the data into k “folds” $f(i)$, that is, **random non-overlapping samples** of the data of size n/k . Then:
- ▶ **For each fold i :**
 - ▶ Call $X^{-(f(i))}$ the “training” dataset and $X^{(f(i))}$ the “test” dataset
 - ▶ Learn parameters $\hat{\theta}_i$ with data $X^{-(f(i))}$
 - ▶ Evaluate $l_i = \text{Loss}(X^{(f(i))} | \hat{\theta}_i)$
- ▶ And report $\frac{1}{n} \sum_{i=1}^k l_i$

How many folds?

- ▶ k -fold CV loses a fraction of the data, whereas LOOCV only loses a constant.
- ▶ This means that (under the assumption that the **true model is not in the model space**) k -fold CV will choose a **simpler model** with less predictive power than was possible.
- ▶ However, smaller k can make the inference more consistent across different data.
- ▶ For **small data**, LOOCV is recommended. For **larger data**, $k = 10$ is often chosen:
 - ▶ **cost**. k defines the minimum number of times you need to run the models. If you can afford to run a model once, you can probably afford 10 times.
 - ▶ **practicality**. If you had only 10% more data you might expect to get the same performance as LOOCV. We frequently lose this amount of data to quality control, etc.

Handling correlation

- ▶ **Correlation** structures can be handled in k-fold CV by **careful sampling**:
 - ▶ a-priori there is a correlation in time or space expected. we can therefore **remove windows**.
 - ▶ the data have some associated covariate, which can be removed en-masse.
 - ▶ empirical correlation structures can be used to select a point i and all points correlated with it above some **correlation threshold**.
- ▶ Some of these can be used in other contexts. Examples include:
 - ▶ **block bootstrap**
 - ▶ Using a different definition of a “datapoint” in a leave-one-out context, for example: datapoints are countries instead of countries at timepoints

Reflection

- ▶ You should understand how to:
 - ▶ Define and use a null hypothesis significance test,
 - ▶ Contrast classical and resampling tests, and judge appropriate uses,
 - ▶ Use statistical testing appropriately in projects.

Further reading

- ▶ Classical Testing
 - ▶ Chapter 4 of [Statistical Data Analysis](#) by Glen Cowan
 - ▶ [Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations](#) by Greenland et al
 - ▶ [Andrew Gelman's blog](#) has many examples of statistical testing failures in social science and medicine
- ▶ Modern Testing
 - ▶ [Cosma Shalizi's Modern Regression Lectures](#) (Lectures 26,28)
 - ▶ [Cross Validation and Bootstrap Aggregating](#) on Wikipedia
 - ▶ Chapters 18.7 of [The Elements of Statistical Learning: Data Mining, Inference, and Prediction](#) (Friedman, Hastie and Tibshirani).
- ▶ Cross Validation
 - ▶ Chapters 2.3 and 7.10 of [The Elements of Statistical Learning: Data Mining, Inference, and Prediction](#) (Friedman, Hastie and Tibshirani).
 - ▶ [Cosma Shalizi's Modern Regression Lectures](#) (Lectures 20, 26)

Appendix: How many permutations?

- ▶ The **smallest possible p-value** with N permutations is $1/(N + 1)$. So 999 permutations gives a minimum of 0.001.
- ▶ The **variance** around a chosen threshold, say $p = 0.05$, is determined by the sampling distribution of the Binomial:

$$\text{sd}(p) = \text{sd}(\text{Bin}(N, p)) = \sqrt{\frac{p(1 - p)}{n}}$$

- ▶ p is of course the true unknown probability, not the observed one.
- ▶ But variance is an increasing function of p (for $p < 0.5$)
- ▶ A heuristic rule is: to be 95% confident that $p \leq t$ we need the empirical p-value to be less than $t - 1.96\text{sd}(p = t)$
- ▶ For $N = 999$ and $t = 0.05$, $\text{sd}(p = t) = 0.0135$ and therefore $p < 0.036$
- ▶ A similar calculation shows $N = 999$ wouldn't be enough to be sure we were less than 0.005.
- ▶ This is conservative... only if the distribution is Normal...(!)
Plot the distribution of T !